

----- Gold Price Prediction Using LSTM Neural Network -----

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Project Type: Machine Learning Project using LSTM

Tools Used: Python, NumPy, Pandas, yfinance(Yahoo Finance), Scikit-learn, TensorFlow/Keras, Matplotlib.

Abstract

This project aims to predict gold prices using a Long Short-Term Memory (LSTM) neural network. Historical gold price data was collected from Yahoo Finance using the ticker symbol "GC=F". The model uses the past 60 days of data to predict the next 7 days. Data preprocessing techniques such as normalization and sequence creation were applied before training the model. The performance of the model was evaluated using Mean Absolute Error (MAE). The results show that the model can capture the overall trend of gold prices, but accuracy decreases when predicting further into the future.

Introduction

Gold price prediction is important in financial markets because it helps investors make better decisions. Gold prices are time-series data, which means past values affect future values. Simple models cannot capture complex patterns in such data. Therefore, LSTM is used because it can learn patterns over time. In this project, an LSTM model is used to predict future gold prices based on past data.

1 .Dataset :

The dataset used in this project was taken from Yahoo Finance using the ticker symbol “GC=F”, which represents gold prices.

- **Data source: Yahoo Finance**
- **Feature used: Closing price**
- **Time period: From 2010 to present**
- **Unit: USD per troy ounce**

This dataset provides daily gold prices, which are used for training and testing the model.

2 . Data Preprocessing :

Before training the model, the data was prepared using the following steps:

- Removed missing values from the dataset
- Split the data into 80% training and 20% testing
- Scaled the data between 0 and 1 using MinMaxScaler
- Avoided data leakage by fitting the scaler only on training data
- Created sequences using a sliding window method:
 - Input: Past 60 days
 - Output: Next 7 days

These steps help the model learn patterns effectively from the data.

3 .Model Architecture :

In this project, an LSTM (Long Short-Term Memory) model is used for prediction. LSTM is suitable for time-series data because it can learn patterns over time.

The model structure is as follows:

- First LSTM layer with 64 units (returns sequences)
- Dropout layer to reduce overfitting
- Second LSTM layer with 32 units
- Dense layer with 7 outputs (for 7-day prediction)

The model uses the Adam optimizer and Mean Squared Error (MSE) as the loss function.

4 Training Process :

The model was trained using the prepared dataset.

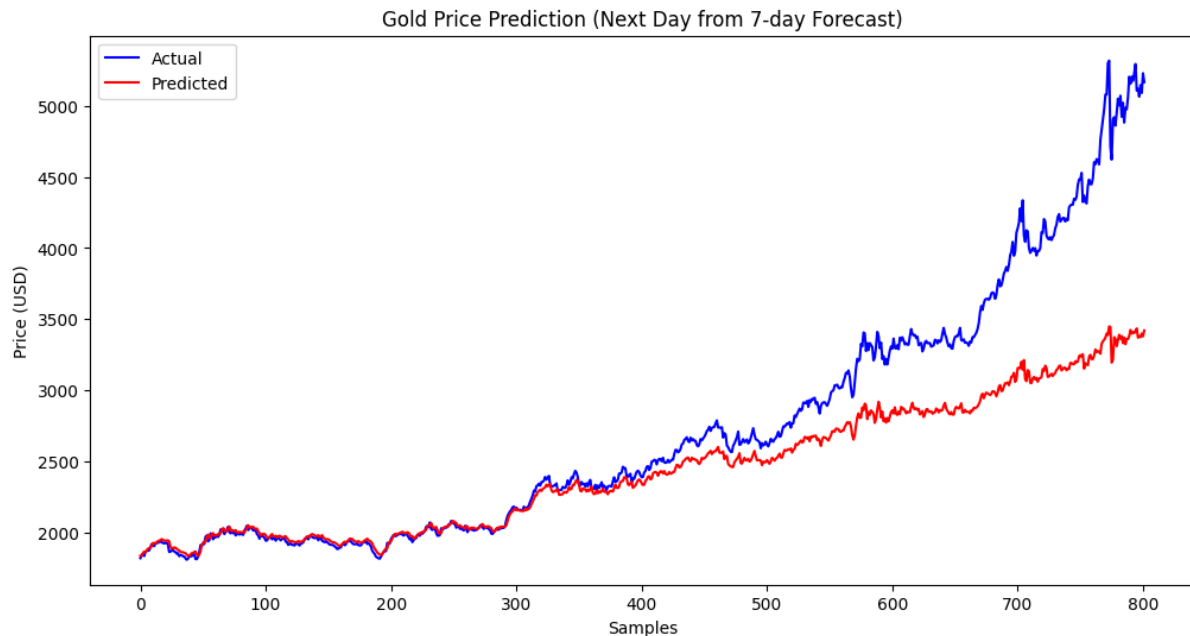
- Training data: 80%
- Testing data: 20%
- Batch size: 32
- Epochs: up to 150

EarlyStopping was used to stop training when the model stopped improving. This helps prevent overfitting.

Validation data was also used during training to check how well the model performs on unseen data.

5 .Results and Analysis :

The model was able to learn patterns from the gold price data and make predictions.



- The model follows the overall trend of gold prices
- Predicted values are smoother than actual values
- The model sometimes underestimates sudden price increases
- Prediction accuracy decreases when predicting multiple days ahead

The performance of the model was evaluated using Mean Absolute Error (MAE), which shows the average difference between predicted and actual prices.

The model sometimes underestimates sudden price changes due to the use of limited features, as only the closing price was used.

Challenges Faced :

During this project, several challenges were faced:

- Data leakage occurred during scaling, which was later fixed.
 - The model showed high error when predicting 30 days ahead.
 - Reducing prediction to 7 days improved performance.
 - The model struggled to capture sudden spikes in price.
 - Results changed slightly each time due to randomness in training.
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Conclusion

This project demonstrates how LSTM can be used for time-series prediction of gold prices. The model is able to capture general trends and provide reasonable predictions based on past data.

Although the model has some limitations, it provides a good foundation for future improvements. Adding more features and improving the model can lead to better accuracy.

References

- Yahoo Finance – Gold Price Data (GC=F)
- NumPy and Pandas documentation
- TensorFlow/Keras documentation
- Scikit-learn documentation
- Matplotlib documentation